

# Droplet Size Evolution in Clouds

## Microphysics Simulation using PySDM

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May 2026

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# Project Objectives



- Simulate the lifecycle of cloud droplets from **nucleation** to **rainfall**.
- Analyze the evolution of the **Droplet Size Distribution (DSD)**.
- Study the efficiency of the **Super-Droplet Method (SDM)**.
- Implementation using the PySDM Python library.

# Core Microphysical Processes



## 1. Nucleation

Activation of aerosol particles into small water droplets.

## 2. Condensation

Growth of droplets due to water vapor diffusion in supersaturated air.

## 3. Collision-Coalescence

Stochastic process where droplets collide and merge into larger rain drops.

# Structure of a Super-Droplet



In the SDM, each computational particle (Super-Droplet) is characterized by a set of attributes:

## Attributes of $i$ -th Super-Droplet

Vector:  $\mathbf{a}_i = (\xi_i, x_i, r_i, m_{sol,i}, v_i \dots)$

- $\xi_i$  (**Multiplicity**): Number of real droplets per super-droplet.
- $x_i$  (**Position**): The position coordinate of the droplet.
- $r_i$  (**Radius**): The physical radius of the droplets in the group.
- $m_{sol,i}$  (**Mass**): Mass of the soluble substance (aerosol) in the droplet.
- $v_i$  (**Velocity**): Velocity of the droplet.

# Motion of Super-Droplets



The trajectory of each super-droplet is tracked in a Lagrangian framework using the time step  $\Delta t_m$ .

## Position Update

The new position is calculated using the standard Euler scheme:

$$\mathbf{x}_i(t + \Delta t_m) = \mathbf{x}_i(t) + \Delta t_m \mathbf{v}_i(t)$$

- **Terminal Velocity:** The velocity  $\mathbf{v}_i(t)$  depends on the local wind and the settling velocity (drag/gravity).
- **Wind Interpolation:** Since wind velocity  $\mathbf{U}$  is defined on a regular grid  $\mathbf{x}_{lmn}$ , we calculate the wind at the droplet's exact position  $\mathbf{x}_i(t)$  via **linear interpolation**:

$$\mathbf{U}_i^* = \text{interp}(\mathbf{U}(\mathbf{x}_{lmn}, t), \mathbf{x}_i(t))$$

# Governing Equation: Condensation



The growth of a single droplet of radius  $r$  is described by Maxwell's equation:

$$r \frac{dr}{dt} = \frac{S - 1}{F_k + F_d} \quad (1)$$

- $S$ : Saturation ratio
- $F_k, F_d$ : Thermodynamic terms (thermal conduction & vapor diffusion)

# Pairwise Collision Probability



The probability  $P_{jk}$  that a pair of super-droplets  $(j, k)$  will collide within time step  $\Delta t$  is:

$$P_{jk} = K(r_j, r_k) \cdot \frac{\max(\xi_j, \xi_k)}{\Delta V} \cdot \Delta t \quad (2)$$

- $K(r_j, r_k)$ : Collision kernel (determines physics of collision).
- $\Delta V$ : Volume of the grid cell (or parcel).
- $\Delta t$ : Time step.

**Crucial optimization:** The SDM calculates collisions between super-droplets, not between all real droplets, reducing complexity.

# PySDM Framework



- **Language:** Python 3.x
- **Acceleration:** Numba (JIT compilation) or PyCUDA for GPU.

```
import PySDM
# Example: Initializing a OD Parcel Model
model = PySDM.Builder(
    backend=PySDM.backends.Numba,
    n_sd=128
).build()
```

# Simulation Goals



- 1 **DSD Broadening:** Observe how the droplet spectrum widens over time.
- 2 **Precipitation Timing:** Identify the "Time-to-Rain" (formation of 1mm drops).
- 3 **Aerosol Impact:** Compare clean vs. polluted air scenarios.

 PySDM Documentation  
<https://pysdm.github.io/>

 Shima et al. (2009)  
The super-droplet method for the numerical simulation of clouds and precipitation.  
QUARTERLY JOURNAL OF THE ROYAL METEOROLOGICAL SOCIETY